MTH631 - Real Analysis II (Quiz No 1)

Question # 5 of 10 (Start time: 03:06:50 PM, 11 November 2025)

The radius of convergence of $\sum a_n(x-x_0)^n$ is given by

Select the correct option

●	$\frac{1}{R}=\lim_{n\to\infty}\left\lvert\frac{a_{n+1}}{a_n}\right\rvert$ if the limit exists in the extended real number system.
○	$R=\lim_{n\to\infty}\left\lvert\frac{a_{n+1}}{a_n}\right\rvert$ if the limit exists in the extended real number system.

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Question # 4 of 10 (Start time: 03:06:31 PM, 11 November 2025)

The infinite series expansion of $\cos x$ for $-\infty < x < \infty$ is given by

Select the correct option

○	$\sum_{n=0}^{\infty}(-1)^n \frac{x^n}{n!}.$
○	$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(n+1)!}.$
○	None of these
●	$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{2n!}.$

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Question # 3 of 10 (Start time: 03:06:05 PM, 11 November 2025)

Let f be defined by a power series with finite radius of convergence R. Then $\lim_{x\to(x_0+R)^-} f(x) = \sum_{n=0}^{\infty} a_n R^n$ only if

Select the correct option

○	None of these.
○	$\sum_{n=0}^{\infty} b_n R^n$ is uniformly convergent.
○	$\sum_{n=0}^{\infty}(-1)^n a_n R^n$ converges.
●	$\sum_{n=0}^{\infty} a_n R^n$ converges

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Question # 10 of 10 (Start time: 03:08:31 PM, 11 November 2025)

Let f be defined by a power series with finite radius of convergence R. Then $\lim_{x\to(x_0+R)+} f(x) = \sum_{n=0}^{\infty} a_n R^n$ only if

Select the correct option

●	$\sum_{n=0}^{\infty} a_n R^n$ converges.
○	$\sum_{n=0}^{\infty}(-1)^n a_n R^n$ converges
○	$\sum_{n=0}^{\infty} b_n R^n$ is convergent.
○	None of the above

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Question # 1 of 10 (Start time: 03:05:13 PM, 11 November 2025)

Consider the function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Select the correct option

●	the function f is infinitely times differentiable on $(-\infty, \infty)$.
○	$f^{(n)}(0) \neq 0$.

Question # 2 of 10 (Start time: 03:05:29 PM, 11 November 2025)

An infinite series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$, is called a power series in $x - x_0$, where

Select the correct option

○	x_0 is constant.
◉	x_0 and $a_0, a_1, \ldots$ are constants.
○	x_0 and $a_0, a_1, \ldots$ are variables.
○	$a_0, a_1, \ldots$ are constants.

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Question # 9 of 10 (Start time: 03:08:07 PM, 11 November 2025)

The power series expansion of $f(x) = \sin x$ about $x_0 = 0$ is

Select the correct option

○	$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty,$
○	$\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,$
●	$\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty.$
○	$\sin x = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,$

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Question # 8 of 10 (Start time: 03:07:36 PM, 11 November 2025)

The term by term differentiation of the series

$$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty$$

becomes

Select the correct option

○	$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty.$
○	$\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty.$
○	$\sum_{n=20}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty.$
●	$\sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad -\infty < x < \infty$

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Question # 6 of 10 (Start time: 03:07:05 PM, 11 November 2025)

For the power series, define R in the extended real numbers by $\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}$. In particular, $R = 0$ if

Select the correct option

●	$\limsup_{n\to\infty} \|a_n\|^{1/n} = \infty.$
○	$\limsup_{n\to\infty} \|a_n\|^{1/n} = 0.$

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Question # 7 of 10 (Start time: 03:07:21 PM, 11 November 2025)

A power series $f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$ with positive radius of convergence R

Select the correct option

●	has derivatives of all orders in its interval of convergence.
○	has continuous limit.